

Between Models: Counterfactual Frontiers across Causal-Response Experiments

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Abstract

Counterfactual conclusions across structural model families depend on the declared causal-response experiment. A family-entry radius and a sign-reversal radius summarize each experiment. Their gap measures the response-space separation between covering every family and reaching a policy-sign reversal. In a monetary application, the baseline four-family experiment gives a gap of 0.8509, and local parameter neighborhoods preserve all 136 delayed-target signs inside the testing-risk benchmark. Re-estimating causal responses by local projections and refitting structural parameters and shock paths gives gaps of -3.2435 , -3.3159 , and -6.1391 under HAC bandwidths 8, 16, and 28. On a common parameter support, shock profiling changes the HAC-28 gap from 6.2580 to -6.1391 ; covariance replacement accounts for most of the remaining fixed-response movement. Positive LP targets remain below 0.001 in discounted cumulative units. A 1,999-draw calendar bootstrap that re-estimates responses and reprofiles every shock path gives a 95-percent gap interval of $[-10.2187, 2.2214]$ and a 6.35-percent positive-gap frequency. The analysis links experiment-specific response geometry, empirical fibers, and generated-response inference.

JEL codes: C12, C13, C32, C52, E52. *Keywords:* structural counterfactuals, partial identification, factorization, weak identification, generated responses, monetary policy.

1. Introduction

How far must a structural model’s causal response move before its monetary counterfactual changes sign? The answer depends on the response experiment used to compare models. An experiment specifies the estimator, response coordinates, shock normalization, covariance, and evidence blocks. The distance to a counterfactual reversal is measured within each experiment and then compared across experiments.

The experiment determines which transmission paths a structural model must complete before its policy target is evaluated. A shock-fitting rule may place weight on a response component that a profiled local-projection experiment absorbs elsewhere. The covariance may also make a given displacement inexpensive along one response direction and costly along another. Cross-model agreement is therefore indexed by the empirical experiment that defines the response neighborhood.

Two radii summarize experiment ξ . The family-entry radius $c_{\text{entry}}(\xi)$ is the distance needed to include at least one representation from every declared structural family. The sign-reversal radius $c_{\text{sign}}(\xi)$ is the smallest distance attained by a representation with a nonnegative policy target. Their gap $\Delta(\xi) = c_{\text{sign}}(\xi) - c_{\text{entry}}(\xi)$ is positive exactly when one response neighborhood contains every family and preserves the negative counterfactual sign. A collection Ξ supports experiment-stable sign separation when $\inf_{\xi \in \Xi} \Delta(\xi) > 0$. Statistical confidence coverage is a separate object.

The gap is a response-space breakdown measure. Masten and Poirier (2020) define boundaries in assumption space that sustain a conclusion, and Kang and Vasserman (2025) measure the functional-form deviation required to overturn welfare conclusions. Here distance is measured in an estimated causal-response experiment, with the search taken over a union of structural families. Evidence budgets generate a frontier, and block decompositions attribute movements in that frontier to observed responses. Bonhomme and Weidner (2022) calibrate misspecification neighborhoods through specification-test power, while Hansen and Sargent (2001, 2008, 2010) provide the testing-risk scale. For a common-covariance Gaussian comparison, total testing error is $2\Phi(-c/2)$; the benchmark $c = 3.9199$ corresponds to total error 0.05. The extrapolation question connects to policy equivalence in Beraja (2023), sufficient-statistic rule counterfactuals in McKay and Wolf (2023), and short-end-shock extrapolation in Caravello et al. (2026). The object studied here is the response distance to an opposing conclusion when several structural families complete the same evidence.

The monetary application begins with a baseline response experiment on a fixed 4,000-evaluation support. All four families enter at 3.5058 and the first evaluated inflation reversal enters at 4.3567, giving a gap of 0.8509. RR narrative-shock responses and the seventh training quarter move the reversing evaluation beyond the benchmark. An outward-rounded 300-quarter calculation extends the four minimum-distance entry points

to positive-width parameter neighborhoods; a fixed center-fitted shock path keeps each neighborhood inside the benchmark, and all 136 delayed-target upper bounds are negative. These fixed-support and local-continuum comparisons complement the moment sensitivity and informativeness measures of Andrews et al. (2017, 2020).

The causal responses are then reconstructed from 275 dated quarters. A lag-augmented local projection gives a nonsingular fifty-coordinate covariance. After structural parameters and shock paths are refitted, the gaps are -3.2435 , -3.3159 , and -6.1391 under HAC bandwidths 8, 16, and 28. On the same 102 parameter points, the original shock-fitting protocol gives a HAC-28 gap of 6.2580, while LP shock profiling gives -6.1391 . Holding the LP-profiled responses fixed and replacing the covariance moves the gap from -1.0918 to -6.1391 . Positive LP crossings remain below 0.001 in discounted cumulative units. A 1,999-draw calendar bootstrap re-estimates the response experiment, re-solves every constrained shock path, and recomputes both binders. Its 95-percent gap interval is $[-10.2187, 2.2214]$, with a 6.35-percent positive-gap frequency. These sampling statements condition on the evaluated structural support and policy targets.

Empirical fibers organize the policy sets behind the radii. Fiber constancy, linear kernel inclusion, and constant-rank factorization state when observed responses determine a policy feature within and across families. Partial-identification foundations follow Tamer (2010), and Kalouptside et al. (2026) develop counterfactual bounds and uniform inference in dynamic discrete-choice models. The generated-response expansion combines minimum-distance, simulation, and implicit-function arguments (Pakes and Pollard 1989; Gouriéroux et al. 1993). Minima and maxima over evaluated structural responses form directionally differentiable functionals (Fang and Santos 2019). Projection provides an identification-strength fallback, and quotient profiling refines regular target-level sets. The reported diagnostics cover rejection, power, set length, and the frequency of unbounded sets.

The family envelope leads to the radius pair, experiment stability, and evidence increments. Factorization conditions and generated-response inference then connect the response geometry to the monetary design and results. Proofs, inference details, simulation design, and additional application calculations appear in the online appendix.

2. The identification problem

2.1. Structural envelopes and response protocols

Let $f \in \{1, \dots, F\}$ index model families. Family f has an economically feasible parameter domain Θ_f^{econ} and model instances $M_f(\theta)$. The *structural envelope* is the disjoint union

$$(1) \quad \mathfrak{E} = \bigsqcup_{f=1}^F \mathfrak{M}_f, \quad \mathfrak{M}_f = \{M_f(\theta) : \theta \in \Theta_f^{\text{econ}}\}.$$

The disjoint union keeps family identity even when two implementations produce the same reduced-form response. The domain Θ_f^{econ} contains the equilibrium, probability, determinacy, borrowing, and calibration restrictions maintained for family f .

Each family supplies three maps:

$$(2) \quad S_{A,f} : \Theta_f^{\text{econ}} \rightarrow \mathbb{R}^{K_A}, \quad S_{B,f} : \Theta_f^{\text{econ}} \rightarrow \mathbb{R}^{K_B}, \quad T_{p,f} : \Theta_f^{\text{econ}} \rightarrow \mathbb{R}^q.$$

The training response S_A corresponds to the causal evidence used to fit or screen the model. The held-out response S_B is reserved for evaluation or residual calibration. The policy map T_p returns the counterfactual target under policy p . A scalar welfare or cumulative-response target has $q = 1$; a path-valued target has $q > 1$.

ASSUMPTION 1 (Common response protocol). *Across all families, S_A , S_B , and T_p use the same outcome definitions, units, shock normalization, policy experiment, horizon, timing convention, and aggregation rule. The implementation declares an equilibrium selection for every evaluated model. Policy-rate innovations are reported in annualized percentage points and enter the quarterly model interface after division by four. Thus 25 basis points equals 0.25 annualized percentage point, and coefficients that combine responses to this basis intervention are dimensionless.*

Empirical equivalence is defined under the common intervention in Assumption 1. Families may still have different state variables and parameter dimensions because the envelope is a disjoint union. The common codomains in (2) place their responses in the same comparison space.

For a model written as $F_f(z, p; \theta) = 0$ with observation map $\Phi_f(z, \theta, p)$, local response derivatives are

$$(3) \quad D_\theta \Phi_f = \Phi_{f,\theta} - \Phi_{f,z} F_{f,z}^{-1} F_{f,\theta}, \quad D_p \Phi_f = \Phi_{f,p} - \Phi_{f,z} F_{f,z}^{-1} F_{f,p},$$

whenever the selected equilibrium has nonsingular $F_{f,z}$. These derivatives enter both the factorization analysis and generated-response inference.

2.2. Empirical fibers and policy sets

DEFINITION 1 (Empirical fiber and policy identified set). For $s \in \mathbb{R}^{K_A}$, define

$$(4) \quad \mathfrak{F}_A(s; \mathfrak{E}) = \{M \in \mathfrak{E} : S_A(M) = s\}, \quad \mathcal{I}_p(s; \mathfrak{E}) = \text{cl}\{T_p(M) : M \in \mathfrak{F}_A(s; \mathfrak{E})\}.$$

The target is point identified relative to $\mathfrak{E}$ at s when the fiber is nonempty and $\mathcal{I}_p(s; \mathfrak{E})$ is a singleton.

The closure includes limiting model sequences at the boundary of a family domain. An empty training fiber receives empirical-compatibility failure status. Identification status is assigned to a nonempty fiber.

DEFINITION 2 (Policy-feature identified set). For a measurable feature $g : \mathbb{R}^d \rightarrow \mathcal{G}$, define

$$(5) \quad \mathcal{I}_{g,p}(s; \mathfrak{E}) = \text{cl}\{g\{T_p(M)\} : M \in \mathfrak{F}_A(s; \mathfrak{E})\}.$$

The feature is identified relative to $\mathfrak{E}$ at s when the fiber is nonempty and $\mathcal{I}_{g,p}(s; \mathfrak{E})$ is a singleton. For componentwise sign identification, $g(t) = (\text{sgn } t_1, \dots, \text{sgn } t_q)$.

Definition 2 separates equality of target values from equality of a policy feature. The monetary application studies the componentwise sign and reports the value image to measure dispersion in magnitudes.

PROPOSITION 1 (Three logically distinct failures). For a declared envelope $\mathfrak{E}$, training response $s_A^\dagger$, held-out response $s_B^\dagger$, and policy p , consider:

1. coverage failure:

$$\mathfrak{F}_A(s_A^\dagger; \mathfrak{E}) = \emptyset;$$

2. policy-identification failure conditional on coverage:

$$\mathfrak{F}_A(s_A^\dagger; \mathfrak{E}) \neq \emptyset, \quad \text{diam } \mathcal{I}_p(s_A^\dagger; \mathfrak{E}) > 0;$$

3. held-out relation failure:

$$s_B^\dagger \notin \{S_B(M) : M \in \mathfrak{F}_A(s_A^\dagger; \mathfrak{E})\}.$$

Coverage failure and conditional policy-identification failure are mutually exclusive. Held-out relation failure accompanies coverage failure and can also occur with either policy identification or policy-identification failure when the training fiber is nonempty.

The first asks whether the declared model envelope contains the empirical object. The second asks whether the observed block is sufficient for the policy target inside that envelope. The third asks whether the same fitted relation predicts a reserved observable

block. In finite samples, projected-moment confidence procedures in Section 5 and Online Appendix D cover the population object in Definition 1.

3. Testing-risk fibers and sign separation

The economic domains in Section 2 define the structural envelope. Statistical distance from the estimated causal response determines the thickness of its empirical fiber. Model admissibility is therefore governed by Θ_f^{econ} , while sampling resolution is measured in response space.

3.1. Thickened empirical fibers

Let $s_A^\dagger \in \mathbb{R}^{K_A}$ be the estimated training response and let $\Sigma_{A,T}$ be its finite-sample covariance on the retained coordinates. Define

$$(6) \quad d_{f,T}(\theta; s_A^\dagger)^2 = \{S_{A,f}(\theta) - s_A^\dagger\}' \Sigma_{A,T}^{-1} \{S_{A,f}(\theta) - s_A^\dagger\}.$$

For radius c , the family neighborhood, thickened fiber, and policy image are

$$(7) \quad \mathcal{N}_f(s_A^\dagger; c) = \{\theta \in \Theta_f^{\text{econ}} : d_{f,T}(\theta; s_A^\dagger) \leq c\},$$

$$(8) \quad \mathfrak{F}_A(s_A^\dagger; c) = \bigsqcup_f \{f\} \times \mathcal{N}_f(s_A^\dagger; c),$$

$$(9) \quad \mathcal{I}_p(s_A^\dagger; c) = \text{cl}\{T_{p,f}(\theta) : (f, \theta) \in \mathfrak{F}_A(s_A^\dagger; c)\}.$$

The zero-radius construction reduces to the exact fiber when $\Sigma_{A,T}$ is positive definite. A positive radius gives a set-valued response compatibility relation indexed by testing risk.

For local geometry, compare two nearby model responses with $\Sigma_{A,T} = \Omega_{A,f}/T$. At a reference θ_f^0 , let $D_f = DS_{A,f}(\theta_f^0)$ and define $H_f = D_f' \Omega_{A,f}^{-1} D_f$.

PROPOSITION 2 (Local response metric). For $\theta = \theta_f^0 + h/\sqrt{T}$,

$$(10) \quad \|S_{A,f}(\theta) - S_{A,f}(\theta_f^0)\|_{\Sigma_{A,T}^{-1}}^2 = h' H_f h + o(1).$$

Under a smooth locally invertible reparameterization $\theta = \varphi(\eta)$,

$$(11) \quad H_{f,\eta} = D\varphi(\eta_0)' H_{f,\theta} D\varphi(\eta_0), \quad \ker H_f = \ker D_f.$$

The local zero-distance directions coincide with the empirical-response kernel. Their policy image $C_{f,p} \ker D_f$ connects the testing-risk geometry to the differential factorization in Section 4.

3.2. Testing affinity and the radius path

For probability laws P_0 and P_1 , define the symmetric total testing risk

$$(12) \quad \tau(P_0, P_1) = \inf_{0 \leq \phi \leq 1} \{\mathbb{E}_{P_0} \phi + \mathbb{E}_{P_1} (1 - \phi)\} = 1 - \|P_0 - P_1\|_{\text{TV}}.$$

PROPOSITION 3 (Gaussian testing-risk calibration). *For two Gaussian experiments with common covariance and Mahalanobis mean distance c ,*

$$(13) \quad \tau(c) = 2\Phi(-c/2), \quad p_{\text{HS}}(c) = \frac{\tau(c)}{2} = \Phi(-c/2), \quad c = -2\Phi^{-1}\{\tau(c)/2\}.$$

Detection-error calibration follows Hansen and Sargent (2001, 2008, 2010); its connection to misspecification neighborhoods and test power is developed by Bonhomme and Weidner (2022). The quantity p_{HS} is the equal-prior detection-error probability. The chosen benchmark uses total testing risk $\tau_{\text{det}} = 0.05$, hence $p_{\text{HS}} = 0.025$ and

$$(14) \quad c_{0.05} = -2\Phi^{-1}(0.025) = 3.9199.$$

Along the optimal Gaussian discriminant, this distance places the facing endpoints of the two 95 percent intervals at the same point. The application reports the full path from $\tau_{\text{det}} = 0.10$ through 0.02.

The scalar sign-reversal radius is a response-space breakdown point in the sense of Masten and Poirier (2020), with a minimum-deviation interpretation related to Kang and Vasserman (2025). The family-entry radius measures representation of structural families. Confidence coverage concerns the sampling procedures in Section 5.

For the radius characterization, take the economic domains to be the closed sets $\Theta_f^{\text{econ}} = \bar{\Theta}_f$. For scalar target coordinates $j \in \mathcal{J}$, define

$$(15) \quad c_{\text{entry}} = \max_f \inf_{\theta \in \Theta_f^{\text{econ}}} d_{f,T}(\theta; s_A^\dagger), \quad U_j(c) = \sup_{(f,\theta) \in \mathfrak{F}_A(s_A^\dagger; c)} T_{j,f}(\theta),$$

$$(16) \quad \delta_{f,j}^{\text{sign}} = \inf_{\theta \in \bar{\Theta}_f: T_{j,f}(\theta) \geq 0} d_{f,T}(\theta; s_A^\dagger), \quad c_{\text{sign}} = \min_{f,j} \delta_{f,j}^{\text{sign}},$$

with the extended-real convention $\inf \emptyset = +\infty$.

PROPOSITION 4 (Family-entry and sign-reversal radii). *Suppose every $\bar{\Theta}_f$ is compact and the response and target maps are continuous. Then $U_j(c)$ is weakly increasing in c . If $c \geq c_{\text{entry}}$, every coordinate of every policy target in $\mathfrak{F}_A(s_A^\dagger; c)$ is negative exactly when $c < c_{\text{sign}}$. Hence family entry and a common negative sign hold on*

$$(17) \quad [c_{\text{entry}}, c_{\text{sign}}).$$

COROLLARY 1 (Persistent interventions). *If Proposition 4 gives $T_{j,f}(h; \theta) < 0$ for delays $h = 0, \dots, 16$, then $\sum_{h=0}^{16} \rho^h T_{j,f}(h; \theta) < 0$ for every $\rho \in [0, 1]$.*

3.3. Stability across empirical experiments

An empirical response experiment ξ declares the estimator, sample, response coordinates, normalization, covariance construction, and structural-input restrictions used in the distance. Holding the policy target and structural-family domains fixed, write its radius pair as $\{c_{\text{entry}}(\xi), c_{\text{sign}}(\xi)\}$ and define

$$(18) \quad \Delta(\xi) = c_{\text{sign}}(\xi) - c_{\text{entry}}(\xi).$$

For a prespecified collection Ξ of response experiments, define the lower experiment gap

$$(19) \quad \underline{\Delta}(\Xi) = \inf_{\xi \in \Xi} \Delta(\xi).$$

PROPOSITION 5 (Experiment-stable sign separation). *Under the conditions of Proposition 4 for every $\xi \in \Xi$, experiment ξ admits an all-family common-negative-sign interval exactly when $\Delta(\xi) > 0$. Every experiment in Ξ admits such an interval when $\underline{\Delta}(\Xi) > 0$. An experiment $\xi_0 \in \Xi$ with $\Delta(\xi_0) \leq 0$ rules out experiment-stable sign separation over Ξ .*

The comparison keeps the structural target fixed while changing the statistical experiment used to assess response compatibility. Each radius retains its own testing-risk scale through Proposition 3. The application reports the radius pair and gap for the baseline response convention and for three raw-data LP covariance constructions.

3.4. Which evidence moves the frontier?

For a subset J of response coordinates, write $r_J(M) = S_J(M) - s_J^\dagger$ and $d_J(M)^2 = r_J(M)' \Sigma_{JJ}^{-1} r_J(M)$. Holding the covariance and model support fixed, define the feature-specific frontier

$$(20) \quad c_j^*(J; \mathfrak{E}) = \inf_{M \in \mathfrak{E}: T_j(M) \geq 0} d_J(M).$$

Thus $c_{\text{sign}} = \min_j c_j^*(A; \mathfrak{E})$ under the same compact economic domains as (16). A superscript $\mathcal{D}$ denotes restriction to an evaluated support.

LEMMA 1 (Evidence increments). *Partition a response block into J and K , and suppose its covariance is positive definite. With $\Sigma_{K|J} = \Sigma_{KK} - \Sigma_{KJ} \Sigma_{JJ}^{-1} \Sigma_{JK}$,*

$$(21) \quad d_{J \cup K}(M)^2 = d_J(M)^2 + \left\| r_K(M) - \Sigma_{KJ} \Sigma_{JJ}^{-1} r_J(M) \right\|_{\Sigma_{K|J}^{-1}}^2.$$

Consequently $\mathfrak{F}_{J \cup K}(s^\dagger; c) \subseteq \mathfrak{F}_J(s^\dagger; c)$ and $c_j^*(J; \mathfrak{E}) \leq c_j^*(J \cup K; \mathfrak{E})$ for a fixed envelope and target. The same conclusions hold on a fixed evaluated support.

The identity is the partitioned Gaussian quadratic-form decomposition. Its use here attributes movement in a nonlocal reversal boundary to response evidence, alongside the local moment-informativeness questions of Andrews et al. (2017, 2020). The second term measures the residual contribution of the added observations conditional on the retained ones. An ablation uses Σ_{JJ} , the covariance of the selected experiment. Re-estimating the parameter support, shock normalization, or covariance defines a further experiment and receives its own report.

3.5. Continuous numerical certificates

Let $\mathcal{D}_f(c)$ index evaluated joint response–policy pairs inside the radius and define their convex response completion

$$(22) \quad \mathcal{C}_f(c) = \left\{ \sum_{i \in \mathcal{D}_f(c)} \lambda_i \begin{pmatrix} S_{A,f}(\theta_{f,i}) \\ T_{p,f}(\theta_{f,i}) \end{pmatrix} : \lambda_i \geq 0, \sum_i \lambda_i = 1 \right\}.$$

PROPOSITION 6 (Convex-completion certificate). *Every empirical-response component of $\mathcal{C}_f(c)$ lies inside the Mahalanobis ball. For target coordinate j ,*

$$(23) \quad \inf_{\mathcal{C}_f(c)} T_{p,j} = \min_{i \in \mathcal{D}_f(c)} T_{p,j}(\theta_{f,i}), \quad \sup_{\mathcal{C}_f(c)} T_{p,j} = \max_{i \in \mathcal{D}_f(c)} T_{p,j}(\theta_{f,i}).$$

Thus a vertex margin $m_j = -\max_i T_{p,j}(\theta_{f,i}) > 0$ certifies $T_{p,j} \leq -m_j$ over this continuous completion.

The convex completion concerns mixtures of evaluated joint response–policy pairs. A native nonlinear parameter-envelope certificate uses a covering or interval argument.

PROPOSITION 7 (Validated covering certificate). *Let $\mathcal{G}_f(c)$ be an ε_f -net of $\mathcal{N}_f(s_A^\dagger; c)$, let $L_{f,j}$ bound the Lipschitz modulus of $T_{j,f}$, and let $e_{f,j}$ bound numerical evaluation error. Then*

$$(24) \quad U_j(c) \leq \max_f \left\{ \max_{\theta \in \mathcal{G}_f(c)} \widehat{T}_{j,f}(\theta) + L_{f,j} \varepsilon_f + e_{f,j} \right\}.$$

A negative right-hand side certifies the sign on the native nonlinear envelope.

3.6. Expansion and approximation

THEOREM 1 (Envelope expansion). *If $\mathfrak{E}_1 \subseteq \mathfrak{E}_2$, then for every s and p ,*

$$(25) \quad \mathcal{I}_p(s; \mathfrak{E}_1) \subseteq \mathcal{I}_p(s; \mathfrak{E}_2).$$

For a declared extension envelope $\mathfrak{G} \supseteq \mathfrak{E}$, a nonempty baseline policy set, and symmetric positive definite Σ_A , define

$$(26) \quad \rho_p(\Delta; \mathfrak{G}) = \inf_{M \in \mathfrak{G}: \text{dist}\{T_p(M), \mathcal{I}_p(s_A^\dagger; \mathfrak{E})\} \geq \Delta} \|S_A(M) - s_A^\dagger\|_{\Sigma_A^{-1}}.$$

The curve gives the empirical distance to an admitted extension that moves the policy set by at least Δ .

Let D collect empirical-response secants and let C_p collect policy-response secants. A reduced-rank map fitted to (D, C_p) represents the evaluated response relation. Its nonlinear approximation error at radius c is

$$(27) \quad \epsilon_{\text{sec}}(c, p) = \max_f \sup_{\theta \in \mathcal{N}_f(s_A^\dagger; c)} \|T_{p,f}(\theta) - \lambda_p\{S_{A,f}(\theta)\}\|_{W_p}.$$

The numerical allowance $\epsilon_{\text{num}}(c, p)$ combines solver, discretization, optimization, and simulation errors. Sampling uncertainty enters the inference procedure in Section 5.

4. Identification and factorization

Population identification requires the policy target to be constant on empirical fibers. At a model instance, kernel inclusion gives the corresponding first-order condition. Under constant rank, this differential restriction integrates over a neighborhood, and agreement on overlaps joins the maps from different families.

4.1. Global factorization on the envelope

THEOREM 2 (Identification as factorization). *For a stated envelope $\mathfrak{E}$, the following are equivalent:*

1. $T_p(M) = T_p(M')$ whenever $S_A(M) = S_A(M')$;
2. there is a unique set map $\lambda_p : S_A(\mathfrak{E}) \rightarrow \mathbb{R}^q$ such that

$$(28) \quad T_p = \lambda_p \circ S_A.$$

Under either condition, $\mathcal{I}_p(s; \mathfrak{E})$ is a singleton for every $s \in S_A(\mathfrak{E})$.

The proof defines $\lambda_p(s)$ as the common policy value on the fiber $S_A^{-1}(s)$. Thus a policy answer factors through the empirical response exactly when every empirically equivalent model gives the same answer. Online Appendix A gives the proof and explains how closure enters the identified set.

4.2. Pointwise and local factorization within a family

Fix a family f and an interior point θ_0 . Write

$$D_f = DS_{A,f}(\theta_0), \quad C_{f,p} = DT_{p,f}(\theta_0).$$

The empirically invisible tangent directions are $\ker D_f$. Their policy image is $C_{f,p} \ker D_f$.

THEOREM 3 (Pointwise differential factorization). *The following are equivalent:*

1. $\ker D_f \subseteq \ker C_{f,p}$;
2. *there is a unique linear map $\Lambda_{f,p,\theta_0} : \text{Im } D_f \rightarrow \mathbb{R}^q$ such that*

$$(29) \quad C_{f,p} = \Lambda_{f,p,\theta_0} D_f.$$

The factor is constructed by $\Lambda_{f,p,\theta_0}(D_f h) = C_{f,p} h$. Kernel inclusion is exactly the condition that makes this definition independent of the chosen preimage. Hence

$$(30) \quad T_{p,f}(\theta_0 + h) = T_{p,f}(\theta_0) + \Lambda_{f,p,\theta_0} D_f h + o(\|h\|).$$

Kernel failure supplies a model perturbation h with $D_f h = 0$ and $C_{f,p} h \neq 0$. Online Appendix A proves existence, linearity, and uniqueness of the factor map.

Neighborhood factorization uses constant rank and neighborhood-wide kernel inclusion. These assumptions produce a quotient chart for the following result.

THEOREM 4 (Local structural factorization). *Suppose $S_{A,f}$ and $T_{p,f}$ are continuously differentiable on an open neighborhood U of θ_0 , $\text{rank } DS_{A,f}(\theta) = r_f$ for every $\theta \in U$, and*

$$(31) \quad \ker DS_{A,f}(\theta) \subseteq \ker DT_{p,f}(\theta) \quad \text{for every } \theta \in U.$$

Then there is a neighborhood $U_0 \subseteq U$ and a unique continuously differentiable map $\lambda_{f,p} : S_{A,f}(U_0) \rightarrow \mathbb{R}^q$ satisfying

$$(32) \quad T_{p,f}|_{U_0} = \lambda_{f,p} \circ S_{A,f}|_{U_0}.$$

The image is endowed with the local embedded-submanifold structure from the constant-rank theorem.

In constant-rank coordinates the training map is $(x, z) \mapsto x$. The vertical coordinate z indexes the local empirical fiber. Condition (31) makes the derivative of the policy target with respect to z vanish, so the target depends on x on a product neighborhood with connected vertical slices. Online Appendix A gives the coordinate argument.

4.3. Agreement across model families

Theorem 4 supplies within-family factorization. Envelope identification additionally requires agreement among families that produce the same training response.

COROLLARY 2 (Within-family sufficiency and cross-family agreement). *Suppose each family admits a factor map $\lambda_{f,p} : S_{A,f}(\Theta_f^{\text{econ}}) \rightarrow \mathbb{R}^d$ satisfying $T_{p,f} = \lambda_{f,p} \circ S_{A,f}$. Then the envelope identifies the policy target on its empirical image if and only if*

$$(33) \quad \lambda_{f,p}(s) = \lambda_{g,p}(s) \quad \text{for every } s \in S_{A,f}(\Theta_f^{\text{econ}}) \cap S_{A,g}(\Theta_g^{\text{econ}})$$

for all f and g .

Equation (33) states the between-model restriction. A fitted predictor measures approximate agreement on the evaluated design. Agreement on the family-image overlaps is evaluated through family-held-out error and boundary searches over the detectability neighborhoods.

4.4. Approximate factorization and policy visibility

For a candidate response map λ_p , define the nonlinear residual

$$(34) \quad R_{f,p}(\theta) = T_{p,f}(\theta) - \lambda_p\{S_{A,f}(\theta)\}.$$

Let $\|\cdot\|_{W_A}$ and $\|\cdot\|_{W_p}$ be declared norms and define the tolerance set

$$(35) \quad \mathcal{J}_p^{\epsilon_A}(s; \mathfrak{E}) = \text{cl}\{T_p(M) : M \in \mathfrak{E}, \|S_A(M) - s\|_{W_A} \leq \epsilon_A\}.$$

LEMMA 2 (Approximate factorization bound). *Suppose $\epsilon_A, \epsilon_C, L_p \geq 0$,*

$$\sup_{M \in \mathfrak{E}} \|T_p(M) - \lambda_p\{S_A(M)\}\|_{W_p} \leq \epsilon_C,$$

and λ_p is L_p -Lipschitz from the empirical norm to the policy norm on the relevant image. Then

$$(36) \quad \text{diam } \mathcal{J}_p^{\epsilon_A}(s; \mathfrak{E}) \leq 2(\epsilon_C + L_p \epsilon_A).$$

The bound follows from two relation residuals and the Lipschitz image of the distance between two admissible training responses. Finite-dimensional norm continuity carries the pairwise bound to the closure. Nonlinear model solutions determine ϵ_C ; in-sample regression fit is reported separately.

For a scalar target with $C_{f,p} \neq 0$ and a symmetric positive definite covariance Σ_A , define the local identification strength

$$(37) \quad \kappa_{f,p}^2 = \inf_{h: C_{f,p}h=1} h' D_f' \Sigma_A^{-1} D_f h.$$

If $\ker D_f \subseteq \ker C_{f,p}$, finite dimensionality and positive definiteness imply $\kappa_{f,p} > 0$. Under kernel failure, a direction with $D_f h = 0$ and $C_{f,p} h \neq 0$ gives $\kappa_{f,p} = 0$. The scalar $\kappa_{f,p}$ records the training-response movement required for a normalized policy movement. The set $C_{f,p} \ker D_f$ records policy movements along the training-response kernel.

5. Inference with generated model responses

The response maps used in the identification analysis are estimated with uncertainty from empirical moments, model parameters, the intervention path, common external inputs, and simulation. A joint first-order representation carries these sources and their cross-family covariance into a projection procedure. A target-constrained quotient uses the geometry of the tested policy level under the stated tangent and covariance conditions.

The statistical references include Stock and Wright (2000) on weakly identified GMM and Kleibergen (2002) on pivotal tests. Andrews and Guggenberger (2019) studies singular moment covariance, AR and conditional quasi-likelihood-ratio tests, and subvector inference. Projection inference here uses a pointwise test over a nuisance union. The quotient results below are regular profiling refinements under explicit tangent conditions, interpreted alongside Andrews and Cheng (2012); Andrews and Mikusheva (2016). Applying the refinement requires estimated covariance, boundary distances, and optimization diagnostics.

5.1. Primitive conditions for generated responses

The expansion combines minimum-distance estimation, simulation asymptotics, and implicit-function differentiation (Pakes and Pollard 1989; Gouriéroux et al. 1993). Its role here is to carry the shared treatment path and overlapping empirical inputs into every family's response and their off-diagonal covariance blocks.

For family f , let $\hat{g}_{f,T}(\theta)$ be a vector of empirical and calibration moments, let $\widehat{W}_f$ be a minimum-distance weight, and let $\hat{\theta}_f$ minimize the corresponding criterion up to first-order error. Write the equilibrium system as $F_f(z, \theta, q) = 0$, where q is the policy-

treatment path estimated from the same or an overlapping sample. A response entering S_A , S_B , or T_p has the form

$$(38) \quad R_f(\theta, q) = \Phi_f\{z_f(\theta, q), \theta, q\}.$$

ASSUMPTION 2 (Generated-response regularity). *For every family, the following conditions hold uniformly on the declared detectability neighborhoods and over the maintained data-generating class.*

1. *The true coordinate is interior, $\hat{\theta}_f \rightarrow_p \theta_f$ uniformly, and the sample first-order condition has remainder $o_p(T^{-1/2})$.*
2. *With $g_f(\theta) = \mathbb{E}\hat{g}_{f,T}(\theta)$ and $G_f = Dg_f(\theta_f)$, the sample moment is stochastically differentiable and*

$$\sqrt{T}\hat{g}_{f,T}(\theta_f) = T^{-1/2} \sum_{t=1}^T \psi_{g_f,t} + o_p(1)$$

under a uniform central limit theorem.

3. *$\widehat{W}_f \rightarrow_p W_f$ uniformly, W_f is symmetric positive definite, G_f has full column rank, and the smallest eigenvalue of $G'_f W_f G_f$ is bounded away from zero.*
4. *The treatment path and all shared inputs have a joint asymptotic linear representation with the calibration moments; in particular, $\sqrt{T}(\hat{q} - q) = T^{-1/2} \sum_t \psi_{q,t} + o_p(1)$.*
5. *The selected equilibrium has uniformly invertible $F_{f,z}$; F_f and Φ_f have bounded continuous derivatives; solver residuals are $o_p(T^{-1/2})$; and, when the estimand is a continuum model, discretization error is $o(T^{-1/2})$.*
6. *Simulation error is centered and has a joint asymptotic linear representation with the preceding inputs. If S_T is the number of simulation draws, $T/S_T \rightarrow \eta \in [0, \infty)$.*

The full-rank condition in part 3 concerns estimation of the declared finite-dimensional coordinate inside each family. The empirical-response derivative $DS_{A,f}$ can have weak singular values or deficient rank under the identification configurations studied below.

THEOREM 5 (Joint representation of generated responses). *Under Assumption 2,*

$$(39) \quad \sqrt{T}(\hat{\theta}_f - \theta_f) = T^{-1/2} \sum_{t=1}^T \psi_{\theta_f,t} + o_p(1), \quad \psi_{\theta_f,t} = -(G'_f W_f G_f)^{-1} G'_f W_f \psi_{g_f,t},$$

and the influence function of (38) is

$$(40) \quad \psi_{R_f} = R_{f,\theta} \psi_{\theta_f} + R_{f,q} \psi_q + \sqrt{\eta} \psi_{\text{sim},f},$$

where

$$(41) \quad R_{f,\theta} = \Phi_{f,\theta} - \Phi_{f,z} F_{f,z}^{-1} F_{f,\theta}, \quad R_{f,q} = \Phi_{f,q} - \Phi_{f,z} F_{f,z}^{-1} F_{f,q}.$$

Stacking families and common inputs yields a joint influence representation with off-diagonal covariance blocks induced by shared samples, treatment paths, and calibration inputs.

Online Appendix C derives (39) from the minimum-distance first-order condition, applies the implicit-function derivative in (41), and tracks the simulation rate. The joint expansion includes treatment-path and common-input covariance alongside parameter-estimation covariance.

5.2. Projection over structural nuisances

Let the nuisance index be $\zeta = (f, \theta)$ in the disjoint union

$$(42) \quad \mathcal{H}(c) = \bigsqcup_f \{f\} \times \mathcal{N}_f(c).$$

The symbol ζ indexes the model nuisance, and η denotes the simulation ratio in Assumption 2. Define population moments

$$(43) \quad g_A(\zeta) = s_A^\dagger - S_A(\zeta), \quad g_\tau(\zeta, \tau) = T_p(\zeta) - \tau, \quad g_B(\zeta) = s_B^\dagger - S_B(\zeta).$$

Let g_S select a fixed-dimensional stack of these moments, with dimension d_S . For a null value b , set

$$(44) \quad Q_{S,T}(\zeta, b) = T \hat{g}_S(\zeta, b)' \hat{\Omega}_S(\zeta)^{-1} \hat{g}_S(\zeta, b), \quad P_{S,T}(b) = \inf_{\zeta \in \mathcal{H}(c)} Q_{S,T}(\zeta, b).$$

ASSUMPTION 3 (Uniform structural-moment approximation). *The joint influence representation and studentized central limit theorem hold uniformly jointly over data-generating laws P in the maintained null class and $\zeta \in \mathcal{H}(c)$. Each of the finitely many family components of $\mathcal{H}(c)$ is compact and the profiled criterion is measurable. The selected covariance matrices are symmetric positive definite with eigenvalues in a fixed compact subset of $(0, \infty)$, their estimators converge uniformly, and d_S is fixed.*

THEOREM 6 (Uniform projection inference). *Consider a composite null class such that, for every P , there is a compatible $\zeta_P^0 \in \mathcal{H}(c)$ with $g_{S,P}(\zeta_P^0, b_P) = 0$. Under Assumption 3,*

$$(45) \quad \limsup_{T \rightarrow \infty} \sup_P \mathbb{P}_P\{P_{S,T}(b_P) > \chi_{d_S, 1-\alpha}^2\} \leq \alpha.$$

The conclusion covers singular empirical-response Jacobians and sequences along which their positive singular values converge to zero.

The infimum in (44) is bounded above by the pointwise statistic at a compatible model nuisance. Uniform studentization gives the $\chi^2_{d_S}$ upper reference at that point. Coverage of the training relation uses g_A . A confidence set for τ inverts the statistic based on $(g'_A, g'_\tau)'$. The reference retains d_S degrees of freedom across changing rank configurations.

5.3. A regular quotient refinement

At a regular interior point θ_0 in family f , let $J_S(\theta_0)$ be the derivative of the selected population moment with respect to structural coordinates and let $r_S = \text{rank } J_S(\theta_0)$. Constant-rank coordinates (x, z) can be chosen so that $x \in \mathbb{R}^{r_S}$ indexes the quotient direction and z indexes the local population-moment fiber.

ASSUMPTION 4 (Local quotient law). *The null is correctly specified at an interior point θ_0 . Economic and detectability inequalities are inactive in a neighborhood. The rank r_S is locally constant, positive singular values of the quotient Jacobian are bounded away from zero, and the local profiled minimizer is consistent for the true quotient coordinate. On the local moment fiber, the first-order covariance law factors through x , $\Omega_S(x, z) = \tilde{\Omega}_S(x)$, and the local quadratic expansion is uniform.*

THEOREM 7 (Regular quotient-space refinement). *Under Assumptions 3 and 4, the locally profiled regular criterion obeys*

$$(46) \quad P_{S,T}^{\text{reg}}(b_0) \Rightarrow \chi^2_{d_S - r_S}.$$

The quotient chart makes the population moment a function of x . The covariance restriction gives the same factorization for the first-order law. The resulting quadratic form is the squared norm of the projection of a d_S -dimensional standard normal vector onto the orthogonal complement of an r_S -dimensional score. Cells with inactive inequalities, locally constant rank, and covariance factorization use the quotient refinement. Other reported cells use the $\chi^2_{d_S}$ projection procedure.

5.4. Regular profiling on a target level set

For a scalar target and family f , define the tested level set and its tangent-response Jacobian by

$$(47) \quad \mathcal{H}_f(b, c) = \{\theta \in \mathcal{N}_f(s_A^\dagger; c) : T_{p,f}(\theta) = b\}, \quad J_{A|b,f}(\theta) = DS_{A,f}(\theta)B_{f,b}(\theta),$$

where the columns of $B_{f,b}(\theta)$ form an orthonormal basis for $\ker DT_{p,f}(\theta)$. Let $Q_{A,f,T}$ use the training moment g_A and its $d_A \times d_A$ covariance, and set

$$(48) \quad P_{f,T}^{\text{tc}}(b) = \inf_{\theta \in \mathcal{H}_f(b,c)} Q_{A,f,T}(\theta).$$

ASSUMPTION 5 (Target-constrained quotient law). *The tested level set is a local manifold around a compatible interior point. The rank $r_{f,b} = \text{rank } J_{A|b,f}(\theta)$ is locally constant, its positive singular values have a uniform positive lower bound, the covariance law factors along the local training-moment fiber inside the level set, and the profiled minimizer is consistent. Economic and detectability inequalities are inactive on the local chart, and the quadratic expansion is uniform.*

THEOREM 8 (Regular target-level profiling). *Under Assumptions 3 and 5,*

$$(49) \quad P_{f,T}^{\text{tc}}(b_0) \Rightarrow \chi_{d_A - r_{f,b}}^2.$$

The result permits weak singular values of $DS_{A,f}$ in directions normal to the tested policy level set.

Family-specific inversions combine as

$$(50) \quad \mathcal{C}_T^{\text{tc}} = \bigcup_f \{b : P_{f,T}^{\text{tc}}(b) \leq \chi_{d_A - r_{f,b}, 1-\alpha}^2\}.$$

Coverage follows from the component containing a compatible family. Use of the refinement is conditioned on target-constrained Jacobian spectra, distances to inequality boundaries, covariance variation along the level set, and optimizer diagnostics from multiple starts.

6. Monetary-policy design

The application asks whether causal responses to current monetary shocks identify the output and inflation effects of delayed or persistent interventions. The empirical response and four-family model evaluations come from Caravello et al. (2026). Their extrapolations are placed in a cross-family policy image indexed by testing risk. The analysis then develops inference for the generated response stack and states numerical certification conditions. Response blocks, extension families, norms, and evidence specifications are set before the corresponding evaluation outcomes are used.

6.1. Training evidence and the common protocol

The baseline quarterly sample is 1969Q1–2006Q4. The reported reconstruction uses the public series and shock archives described below, with a 2026-08-25 FRED vintage for the current-vintage analysis. The aggregate variables are log real output per capita (FRED A939RX0Q048SBEA), transformed following Hamilton (2018); annualized log-difference inflation in the GDP deflator (FRED GDPDEF); and the federal funds rate (FRED FEDFUNDS). The two treatment series are the Romer and Romer (2004) monetary-policy shock, using the Ramey replication series, and the natural-language shock of Aruoba and Drechsel (2024). Monthly shocks are aggregated to quarters, with missing shock observations set to zero in the baseline design.

The VAR order is AD shock, output, inflation, RR shock, and the policy rate, with two lags, an intercept, and a linear trend. Recursive identification fixes the impact responses of output and inflation to the RR innovation at zero. The covariance assigns variance 10^6 to these two coordinates following the author implementation, and the horizon-8 triangular taper applies before the paper-specific selection and scaling.

Within the baseline 1969Q1–2006Q4 VAR design, let $q \in \{RR, AD\}$ denote the two treatments and let $a_q = 0.25 / \max_{0 \leq j \leq 24} |IRF_i^q(j)|$. Every response for treatment q is multiplied by a_q , so its annualized policy-rate path has a 25-basis-point peak over the first 25 quarters. Output is measured in percent deviations; inflation and the policy-rate input are divided by four at the model interface. The training response stacks output and quarterly inflation over 13 quarters:

$$(51) \quad s_A^\dagger = \begin{pmatrix} IRF_y^{RR}(0:12) \\ IRF_\pi^{RR}(0:12) \\ IRF_y^{AD}(0:12) \\ IRF_\pi^{AD}(0:12) \end{pmatrix} \in \mathbb{R}^{52}.$$

The 52 outcome coordinates and 26 policy-rate coordinates form a 78-dimensional joint input with covariance obtained from the same transformed VAR draws. The response reconstruction implements VAR and lag-augmented LP, following the population-response connection in Plagborg-Møller and Wolf (2021) and the lag-augmentation design of Montiel Olea and Plagborg-Møller (2021). It refits Hamilton preprocessing and peak normalization on the dated raw-data calendar. Alternative shock constructions and post-2006 samples have separate response and covariance requirements.

6.2. The structural envelope

The filtering ablation holds the 4,000 baseline evaluations, fitted two-shock input paths, normalization, and policy responses fixed. Twenty-six evidence specifications select shock blocks, outcome blocks, individual shock–outcome blocks, their complements, and joint horizons zero through k for every $k = 0, \dots, 12$. Each specification uses its principal submatrix of the baseline covariance and the same testing-risk radius. The empty-filter row describes the original estimated support. The experiment measures changes in compatibility within this support; parameter and shock-input re-estimation define a separate sensitivity exercise.

The baseline evaluated support consists of 1,000 retained evaluations per family. Its coordinate hull supplies a bounded search region, with fixed coordinates held at their baseline values. An economic admissibility domain additionally imposes the family’s probability, equilibrium, determinacy, borrowing, and calibration restrictions. The reported parameter searches use the coordinate hull and the finite-horizon solver selection and are interpreted on that computational domain.

The four baseline families are summarized in Table 1. Their common interface returns S_A , the candidate held-out blocks S_B , and policy responses T_p under the protocol in Assumption 1.

TABLE 1. Baseline model families

Family	Defining margin	Family-specific calibration content
RANK-RE	Representative agent; rational expectations	Aggregate monetary responses and standard nominal/real rigidity moments
HANK-RE	Household risk and portfolio heterogeneity; rational expectations	Common aggregate moments plus liquid wealth, debt exposure, MPC, and idiosyncratic-risk moments
RANK-CD	Representative agent; cognitive discounting in price and wage setting	RANK moments plus expectation-response moments when included in the evidence design
HANK-CD	Household heterogeneity and cognitive discounting	HANK distributional moments plus expectation-response moments

The RANK and HANK benchmarks follow the monetary-transmission specifications of Christiano et al. (2005) and Kaplan et al. (2018). HANK computation uses the sequence-space methods of Auclert et al. (2021); cognitive discounting follows Gabaix (2020). Parameter domains and equilibrium selections are described in Online Appendix E.

The baseline cognitive-discounting calibration is $m = 0.65$. The value $m = 1$ corresponds to rational expectations in this implementation. Under weak separation in the calibration experiment, m enters the reported sensitivity coordinates. Each family defines Θ_f^{econ} from

its equilibrium and calibration restrictions. The response-detectability distance centers $S_{A,f}(\theta)$ at the estimated 52-vector $s_A^\dagger$ and uses the baseline finite-sample covariance in (6).

The benchmark sets symmetric total testing risk to five percent. Following (14), the corresponding Hansen–Sargent detection-error probability is 0.025 and the radius is 3.9199. The reported path is

$$(52) \quad \tau_{\text{det}} \in \{0.10, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02\},$$

$$(53) \quad p_{\text{HS}} = \tau_{\text{det}}/2, \quad c = -2\Phi^{-1}(p_{\text{HS}}).$$

The radius path screens 1,000 retained baseline evaluations per family under a fixed covariance and response design. Separate coordinate and damped Gauss–Newton searches supply nonlinear coverage witnesses. The native-envelope certificate in Proposition 7 additionally requires closed parameter domains, equilibrium and conditioning gates, target-specific interval bounds, and zero unresolved boxes. A family holdout fits the response relation on the complement of that family’s evaluated points.

The extension exercise associated with (26) adjoins one author-calibration point from the financial model of Christiano et al. (2014) to the 4,000 baseline evaluations. The CMR parameter vector and economic equations retain their author values; 24 news states implement the common dated-wedge protocol. Two 25-quarter shock sequences are fitted to the normalized 150-coordinate empirical response before computing the 52-coordinate training distance. The calculation reported here treats this CMR calibration as an evaluated expansion point.

6.3. Policy targets and disagreement geometry

Policy $p = h$ is a one-quarter additive monetary-policy wedge at horizon h , announced at date zero and measured as 25 annualized basis points:

$$(54) \quad u_t^{(h)} = 0.25 \mathbb{1}\{t = h\}, \quad h = 0, 1, \dots, 16,$$

where the solver receives the quarterly counterpart. Each family’s maintained feedback rule remains in force, so the equilibrium interest-rate response is an outcome of the intervention. The common object is the announced wedge sequence. A prescribed equilibrium rate path and a change in feedback-rule coefficients define further policy objects. The corresponding model response is

$$(55) \quad v_f(h; \theta) = \begin{pmatrix} IRF_{y,f}^{(h)}(0:12; \theta) \\ IRF_{\pi,f}^{(h)}(0:12; \theta) \end{pmatrix} \in \mathbb{R}^{26}.$$

The scalar targets are cumulative discounted output and inflation responses,

$$(56) \quad \tau_Y(h; \theta) = \sum_{t=0}^{12} \beta^t IRF_y^{(h)}(t; \theta), \quad \tau_\pi(h; \theta) = \sum_{t=0}^{12} \beta^t IRF_\pi^{(h)}(t; \theta),$$

with baseline $\beta = 0.99$ and sensitivity values 0.95 and 1.

For family f with n_f centered nonlinear design evaluations, let $D_f \in \mathbb{R}^{52 \times n_f}$ collect training-response secants and $V_f(h) \in \mathbb{R}^{26 \times n_f}$ collect policy-response secants. Let $\widehat{\Lambda}_{-f,r}(h) \in \mathbb{R}^{26 \times 52}$ be a rank- r map fitted on families in $\{1, \dots, F\} \setminus \{f\}$, and let $W_v \in \mathbb{R}^{26 \times 26}$ be symmetric positive definite. The family-held-out loss is

$$(57) \quad E_f(r, h) = \left\| W_v^{1/2} \{V_f(h) - \widehat{\Lambda}_{-f,r}(h) D_f\} \right\|_F^2.$$

Families receive equal total weight in $CV(r, h) = \sum_f \omega_f E_f(r, h)$. The selected rank is the smallest rank within one family-clustered standard error of the minimum. Reports label it the finite-design rank and place it beside the population-Jacobian rank and the nonlinear residual in (27).

At every family center and policy horizon, the application computes $D_{A,f} = DS_{A,f}(\theta_f^0)$, $C_{Y,f}(h) = D\tau_{Y,f}(h; \theta_f^0)$, and $C_{\pi,f}(h) = D\tau_{\pi,f}(h; \theta_f^0)$. The output includes $C_{p,f}(h) \ker D_{A,f}$ and $\kappa_{f,p}(h)$ from (37). Cross-family overlap is examined at responses lying in, or statistically compatible with, two family images.

6.4. Additional response evidence

The policy-information block uses the public replication shocks of Jarociński and Karadi (2020) with one- and two-year Treasury-yield responses. The target/path decompositions of Gürkaynak et al. (2005) and Swanson (2021), and the observation protocol of Bauer and Swanson (2023), enter source-availability sensitivities. Survey expectations follow the response design in Del Negro et al. (2023). Together these form the expectations and policy-information block B_E .

The financial block B_F contains commercial-paper funding spreads and Census QFR investment and debt responses by asset-size and industry cells, motivated by Gertler and Karadi (2015) and Ottonello and Winberry (2020). The QFR analysis has an aggregate-cell estimand. The household block B_H uses SCF exposures motivated by Auclert (2019) and CEX consumption responses by homeownership, mortgage-payment status, and liquid-resource group motivated by Cloyne et al. (2020). These blocks provide additional response coordinates for transmission checks and family-held-out prediction.

The evaluation design reserves long horizons, alternating output and inflation coordinates, external policy-information responses, expectation and balance-sheet responses, and one structural family at a time. The external blocks are reported as response estimates

with their uncertainty, while family-held-out exercises evaluate predictive coverage and generated-response loss.

6.5. Estimation, numerical certification, and policy exercises

Each family is estimated by common minimum distance,

$$(58) \quad \hat{\theta}_f = \underset{\theta \in \Theta_f^{\text{econ}}}{\operatorname{argmin}} \{ \hat{\mu}_f - \mu_f(\theta) \}' \widehat{\Omega}_{\mu,f}^{-1} \{ \hat{\mu}_f - \mu_f(\theta) \}.$$

Shared samples, treatment paths, and external moments are stacked before covariance estimation. Numerical accuracy is assessed through equilibrium residuals, horizon refinement, solver-tolerance comparisons, and simulation variation. Independent horizon-200 and horizon-300 solves are compared for the 2,457 compatible vertices, and the resulting numerical discrepancies are reported in Online Appendix E.

Persistent paths are constructed from the same 0.25-percentage-point basis:

$$(59) \quad u_t(\rho) = 0.25\rho^t, \quad t = 0, \dots, 16, \quad \rho \in \{0, 0.50, 0.75, 0.90\},$$

and, in the linear response system,

$$(60) \quad v(\rho) = \sum_{h=0}^{16} \rho^h v(h).$$

The coefficient ρ^h is dimensionless because $v(h)$ already corresponds to a wedge whose annualized unit is 0.25 percentage point. The intervention equals zero after quarter sixteen. Equation (60) therefore describes a truncated persistent intervention; the equilibrium response after quarter sixteen follows the maintained model and policy rule.

The rule exercise sends the synthesized policy-response set to the solver in McKay and Wolf (2023) or Hebden and Winkler (2026). The baseline rule targets output-gap stabilization and is fixed before computation. The sequence is

empirical responses $\longrightarrow$ between-model policy set
 $\longrightarrow$ downstream rule solver $\longrightarrow$ rule-counterfactual set.

For this prespecified exercise, the rule coefficients and solver protocol are held fixed across admissible response inputs. Reports place response-input identification and numerical rule solution in two consecutive stages.

The Monte Carlo varies sample size, the smallest singular value of $D_{A,f}$, the policy loading along its right singular vector, and the covariance between generated responses and derivatives. It compares plug-in Wald, rank-threshold Wald, projection inference, the

regular quotient, and a target-constrained quotient oracle. Reported cells use 5,000 replications and record rejection frequency at a five-percent level, 95 percent confidence-set coverage, median set length, power, unbounded-set frequency, and empty-set frequency. Online Appendix E gives the simulation grid and implementation details.

7. Counterfactual frontiers across causal-response experiments

Frontiers move across empirical experiments as response restrictions and structural fitting change. Each result is reported with its sampling scheme and structural domain. Structural sign separation concerns the policy image of a declared response-compatible set. Sampling sign exclusion concerns a confidence set that propagates the stated estimation uncertainty.

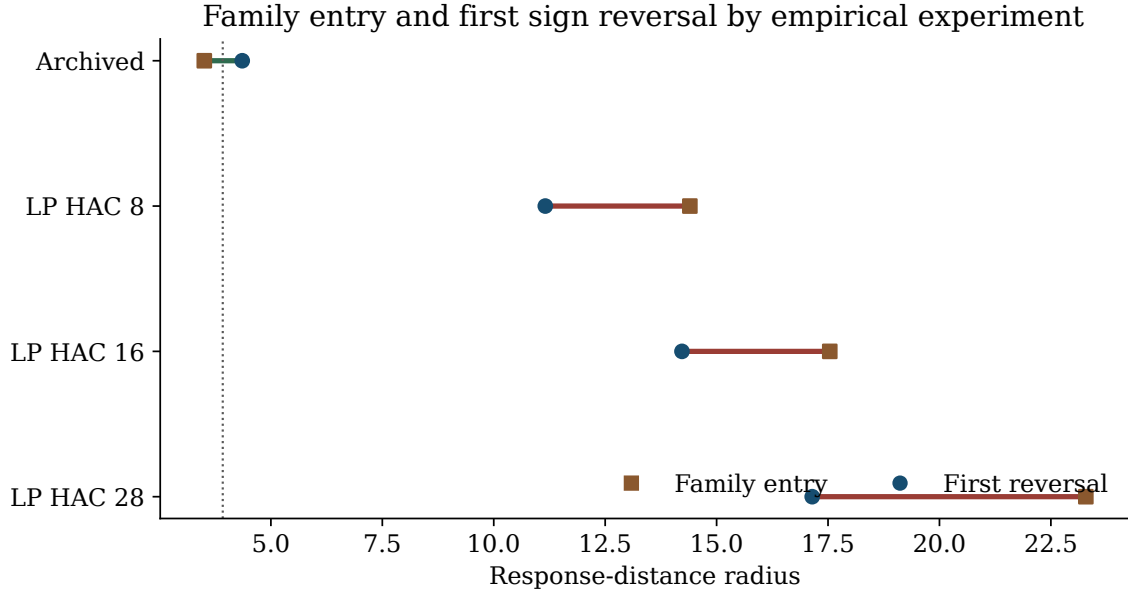
7.1. The experiment-indexed result

The baseline response experiment yields family-entry radius 3.5058 and first evaluated reversal 4.3567, so its evaluated-support gap is 0.8509. Refitting parameters and shock paths to the fifty-coordinate LP experiment yields negative gaps under all three HAC bandwidths: -3.2435 , -3.3159 , and -6.1391 . Figure 1 displays the change in ordering. The four-experiment collection has $\inf_{\xi \in \Xi} \Delta(\xi) < 0$ on the declared evaluated supports.

On a common set of 102 parameter points, the original shock-fitting protocol gives a HAC-28 gap of 6.2580 and LP shock profiling gives -6.1391 . Holding the LP-profiled fitted responses fixed, the baseline active covariance gives gap -1.0918 and the LP HAC-28 covariance gives -6.1391 . The first LP crossings are positive by at most 0.000284 for output and 0.000814 for inflation. Tables 16 and 20 report the decomposition and target-size frontiers.

Sampling statements depend on how the response experiment is resampled. Fixed-support Gaussian directional ranges lie below zero for each LP bandwidth. Calendar block-multiplier ranges include zero after preprocessing, LP estimation, impact identification, and normalization are rerun. Tables 17 and 18 report these two calculations. Each result retains its stated structural support, policy targets, covariance treatment, and profiling regime.

FIGURE 1. Family entry and first sign reversal across empirical experiments



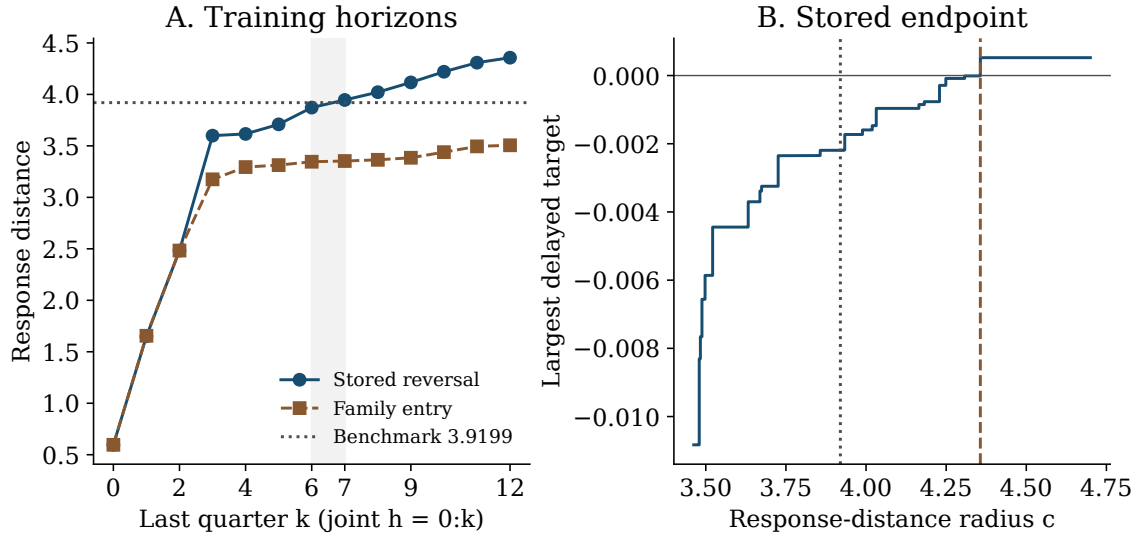
Squares mark the radius at which every family has an evaluated representation; circles mark the first evaluated nonnegative target. Green joins a positive gap and red joins a negative gap. The dotted line is the benchmark radius 3.9199. The baseline and LP supports and covariance constructions are stated separately.

7.2. Which evidence moves the baseline frontier

On the 4,000-evaluation baseline support, the distance needed for coverage of all four families is 3.5058. The first evaluated sign reversal is at 4.3567. All-family evaluated entry and common negative delayed signs hold on the interval $[3.5058, 4.3567)$. The upper boundary corresponds to symmetric total testing risk 0.02938. A benchmark at 3.9199 leaves distance slack 0.4367 inside this interval; the frontier also gives the conclusion for every other radius.

Table 2 holds the structural evaluations fixed and changes the response coordinates used to filter them. With the AD shock alone, all 4,000 evaluations enter and twelve inflation targets admit positive values. With the RR shock alone, 3,813 enter and every inflation endpoint is negative. Combining both shocks retains 2,457 and moves the inflation reversal distance from 4.1579 to 4.3567. Output endpoints are negative across the initial support, including the empty-filter row. This output result describes a restriction of that support; the inflation comparison measures exclusion by the response evidence.

FIGURE 2. Evaluated frontiers over training horizons and response radius



Panel A keeps the 4,000 structural evaluations and fitted inputs fixed. The shaded interval marks the addition of quarter seven. Panel B reports the largest of the 34 delayed target endpoints. Both panels use the baseline covariance and describe structural filtering.

TABLE 2. Evidence and the distance to sign reversal

Evidence	Retained	$c_{\text{entry}}^{\mathcal{D}}$	$c_{\text{sign}}^{\mathcal{D}}$	$Y < 0$	$\pi < 0$
Empty filter	4,000	0.0000	0.0000	17/17	5/17
AD	4,000	1.4476	1.8164	17/17	5/17
RR	3,813	3.0602	4.1579	17/17	17/17
Output, both shocks	3,997	2.5815	2.5815	17/17	5/17
Inflation, both shocks	4,000	2.5263	3.4837	17/17	5/17
Joint horizons 0–3	3,884	3.1750	3.5993	17/17	5/17
Joint horizons 0–6	3,497	3.3459	3.8719	17/17	5/17
Joint horizons 0–7	3,396	3.3529	3.9460	17/17	17/17
Joint horizons 0–12	2,457	3.5058	4.3567	17/17	17/17

The support contains the same 4,000 baseline evaluations in every row. Retained counts and negative-cell counts use $c = 3.9199$, $\beta = 0.99$, and delays zero through sixteen. Each distance uses the principal covariance submatrix of its evidence block. $c_{\text{sign}}^{\mathcal{D}}$ is the first evaluated reversal distance; $c_{\text{entry}}^{\mathcal{D}}$ is the largest family-specific minimum distance. The parameter support, fitted shock inputs, and policy targets remain fixed.

The horizon sequence locates this exclusion within the observed response path. Joint horizons zero–six retain 3,497 evaluations and put the reversing point at distance 3.8719. Adding quarter seven raises its distance to 3.9460, retains 3,396 evaluations, and makes all seventeen inflation endpoints negative. The smallest inflation margin at that step is 0.00008548. Extending the response block through quarter twelve yields margin 0.002193.

Online Appendix E reports all thirteen nested horizon designs and the shock–outcome block deletions.

HANK–RE draw 443 is the only baseline evaluation with a nonnegative delayed target. It supplies twelve delayed inflation reversal cells and four persistent inflation reversal cells. Its delay-three inflation target is 0.00052377. The remaining 22 delayed cells and four persistent output cells have zero reversing evaluations in the evaluated support. These records specify the numerical frontier’s support. Parameter searches supply additional reversal witnesses under the baseline metric: HANK–RE reaches 4.2077 at inflation delay three, and RANK–RE reaches 4.1847 at inflation delay six. The latter target is 0.00011925 after a 300-quarter solve. Each evaluated witness gives an upper bound on the corresponding parameter-domain infimum.

The conditional-distance decomposition in Lemma 1 identifies the empirical directions that separate draw 443. Its AD distance is 1.8164 and its RR distance is 4.1579. Adding RR conditional on AD contributes 15.6813 to squared distance; adding AD conditional on RR contributes 1.6928. Both blocks together give squared distance 4.3567^2 . These increments account for the cross-shock covariance and hold the original parameter support and fitted input paths fixed.

7.3. An executed CMR calibration point

The CMR author-calibration point supplies an output reversal at delay sixteen. Its discounted cumulative output target equals 0.02052. In the first thirteen response quarters, the consumption contribution is -0.12370 and the investment contribution is 0.14422 ; government expenditure is unchanged. This target measures the response before the delayed tightening is implemented. All seventeen inflation targets and the eight reported persistent output and inflation targets are negative. The extension therefore locates a disagreement in the composition and timing of expenditure before an announced future wedge.

The CMR point’s joint training distance is 4.8959. AD evidence admits it at distance 2.4288; RR evidence puts it at 4.6164. Table 3 shows that the AD-filtered expanded support has sixteen negative output cells, and the RR-filtered expanded support has seventeen. The full-evidence benchmark excludes this CMR point. On the 4,001-point expanded support, the output sign frontier becomes 4.8959 and the first joint output–inflation frontier remains 4.3567. These are bounds for the evaluated expansion with a fixed author calibration. Parameter estimation and domain-wide optimization define the remaining CMR-family calculation.

TABLE 3. CMR author-calibration point and evidence filtering

Evidence	CMR distance	Admitted	$Y < 0$	$\pi < 0$
Empty filter	0.0000	1	16/17	5/17
RR	4.6164	0	17/17	17/17
AD	2.4288	1	16/17	5/17
Joint horizons 0–3	3.7651	1	16/17	5/17
Joint horizons 0–7	4.3452	0	17/17	17/17
Joint horizons 0–12	4.8959	0	17/17	17/17

The expanded evaluated support has 4,000 baseline evaluations and one CMR author-calibration point. Negative-cell counts use the union admitted at $c = 3.9199$. The CMR economic parameters are fixed; two 25-quarter monetary-wedge sequences are fitted to the normalized 150-coordinate empirical response. A 24-state news chain implements the delayed interventions under the existing CMR policy rule.

The execution uses the public CMR code with an API port to Dynare 7 and Octave. Its 349-equation author system has maximum static residual 1.29×10^{-12} . The news augmentation has 373 equations and 94 unstable eigenvalues for 94 forward variables. The contemporaneous response agrees with the original author solve within 6.54×10^{-12} per annualized percentage point. Output constructed from expenditure components agrees with accumulated GDP growth within 3.74×10^{-13} . These numerical checks verify the response construction used for the reported CMR point.

7.4. Causal response design

The empirical response vector uses the narrative shock of Romer and Romer (2004) and the text-based shock of Aruoba and Drechsel (2024). Under the maintained shock-exogeneity conditions, the 52 output and inflation responses identify the causal effect of unanticipated policy movements over the first thirteen quarters. The independent response reconstruction reproduces the author OLS impulse responses within 8.66×10^{-11} . Each shock is scaled so that the annualized policy-rate response reaches 25 basis points over horizons zero through twenty-four; the corresponding multipliers are 0.361 for RR and 0.705 for AD.

The event-date design separates the policy component from the information-shock component for 319 announcements. Table 4 reports the response of one- and two-year Treasury yields. At the event date, the four max- t intervals lie above zero. The joint impact tests have p -values 2.82×10^{-18} for the one-year yield and 6.29×10^{-14} for the two-year yield. The five-date policy coefficients remain positive under simultaneous coverage.

TABLE 4. H3 event-date Treasury-yield holdout

Yield and window	MP coefficient [simultaneous interval]	CBI coefficient [simultaneous interval]	Joint p
DGS1, event date	0.574 [0.321, 0.827]	0.878 [0.504, 1.253]	2.82×10^{-18}
DGS1, 5 dates	0.886 [0.468, 1.303]	0.647 [-0.213, 1.507]	4.55×10^{-13}
DGS1, 20 dates	0.904 [-0.071, 1.879]	1.340 [-0.253, 2.933]	1.31×10^{-4}
DGS2, event date	0.650 [0.283, 1.016]	0.861 [0.249, 1.474]	6.29×10^{-14}
DGS2, 5 dates	0.705 [0.242, 1.168]	0.440 [-0.304, 1.184]	3.90×10^{-6}
DGS2, 20 dates	0.526 [-0.411, 1.463]	1.126 [-0.469, 2.721]	0.0244

Each regression uses 319 announcement dates and includes a constant, the median-rotation monetary-policy shock, and the median-rotation central-bank-information shock. Coefficients measure percentage-point yield changes per one-unit shock in the public file. Brackets use the 95 percent max- t interval across 12 slope coefficients from 9,999 event-level Rademacher multiplier draws. Joint p -values use the two-slope HC3 Wald statistic.

The expectation block uses individual Survey of Professional Forecasters submissions from 66 survey quarters. A narrative tightening lowers real-growth forecasts three and four horizons ahead. The RR coefficients are -0.622 and -0.713 , with simultaneous intervals $[-1.198, -0.046]$ and $[-1.294, -0.133]$. A survey-quarter mean specification with Newey–West covariance gives -0.633 and interval $[-1.255, -0.012]$ at horizon three.

TABLE 5. H4E Survey of Professional Forecasters holdout

Forecast outcome	h	RR coefficient	AD coefficient	Responses
Real GDP growth	0	0.21 [-2.86, 3.28]	4.30 [-4.21, 12.80]	2,416
	1	-0.08 [-1.80, 1.65]	1.93 [-3.54, 7.40]	2,416
	2	-0.36 [-1.15, 0.43]	1.48 [-2.65, 5.62]	2,412
	3	-0.62 [-1.20, -0.05]	1.45 [-2.54, 5.43]	2,397
	4	-0.71 [-1.29, -0.13]	1.74 [-2.17, 5.65]	2,338
GDP-price growth	0	0.12 [-1.27, 1.51]	2.94 [-0.94, 6.83]	2,323
	1	0.43 [-0.60, 1.45]	2.06 [-1.25, 5.37]	2,322
	2	0.48 [-0.47, 1.42]	1.63 [-1.83, 5.08]	2,319
	3	0.64 [-0.38, 1.66]	1.16 [-2.11, 4.43]	2,302
	4	0.59 [-0.37, 1.54]	0.94 [-2.44, 4.32]	2,250

Each outcome is the annualized log growth forecast between adjacent SPF level forecasts. Regressions use shocks from the preceding calendar quarter. Brackets give 95-percent intervals simultaneous across 20 slopes, formed from common survey-quarter Rademacher multipliers and the coordinatewise larger forecaster- and calendar-clustered standard error. A survey-quarter mean sensitivity uses Newey–West covariance with lag $h + 1$ and Bonferroni coverage; its RR coefficient for real GDP growth at $h = 3$ is -0.633 with interval $[-1.255, -0.012]$.

The yield and forecast estimates locate two links in the transmission sequence. Policy announcements move short Treasury yields immediately, and the narrative tightening lowers expected real growth at longer forecast horizons. The structural envelope uses the aggregate output and inflation responses for estimation and reserves these market and expectation responses for external validation.

7.5. Testing-risk coverage and radius path

The benchmark total testing risk is 0.05, its Hansen–Sargent detection-error probability is 0.025, and its radius is 3.9199. Table 6 reports the retained baseline evaluations and the best recorded nonlinear coverage witness from the coordinate and damped Gauss–Newton searches.

TABLE 6. Response coverage at total testing risk 0.05

Family	Retained minimum	Retained count	Nonlinear witness	Witness max target
RANK–RE	3.4880	613	3.4280	−0.005974
HANK–RE	3.5058	561	3.4245	−0.007058
RANK–CD	3.4838	664	3.3857	−0.007743
HANK–CD	3.4623	619	3.4089	−0.009090

The radius is 3.9199. Each family supplies 1,000 retained baseline evaluations. The nonlinear-witness column reports the best distance recorded by coordinate search and damped Gauss–Newton refinement; its projected-gradient norms are 4.299, 6.555, 1.675, and 4.700 in row order. Each witness is a feasible coverage point and its 34 delayed-policy targets lie below zero.

Each family has a nonlinear coverage witness with distance from 3.3857 to 3.4280. The projected-gradient norms range from 1.675 to 6.555, so these distances serve as feasible witness values. The 4,000 retained points are 1,000 retained parameter draws per family. Their counts summarize the evaluated design; sampler density governs these counts. At the benchmark, 613 RANK–RE, 561 HANK–RE, 664 RANK–CD, and 619 HANK–CD points enter the response ball.

Table 7 traces the same baseline design over eight testing-risk levels. The radius associated with total risk 0.10 admits zero retained points. All four families have retained coverage by total risk 0.07. The largest policy upper endpoint remains below zero through total risk 0.03 and becomes positive by total risk 0.02.

TABLE 7. Baseline-design policy image over the testing-risk path

Total risk	p_{HS}	Radius	Family counts	Total	Largest policy upper
0.10	0.050	3.2897	0/0/0/0	0	Coverage empty
0.08	0.040	3.5014	1/0/3/8	12	−0.005861
0.07	0.035	3.6238	55/59/64/95	273	−0.004444
0.06	0.030	3.7616	279/289/305/331	1,204	−0.002351
0.05	0.025	3.9199	613/561/664/619	2,457	−0.002193
0.04	0.020	4.1075	872/806/901/820	3,399	−0.000963
0.03	0.015	4.3402	958/947/987/948	3,840	−0.000011
0.02	0.010	4.6527	998/995/1,000/995	3,988	0.002816

Family counts are RANK-RE/HANK-RE/RANK-CD/HANK-CD. The last column is the largest upper endpoint across 34 delayed and eight persistent target cells. Counts depend on the retained-chain design.

The first evaluated sign reversal enters at distance 4.3567, equivalent to total testing risk 0.02938 and Hansen-Sargent error 0.01469. It is HANK-RE draw 443: inflation at delay three equals 0.000524, and the $\rho = 0.90$ inflation response equals 0.002816. The benchmark therefore has a baseline-design sign-radius slack of 0.4367.

7.6. Continuous response completion and local parameter neighborhoods

The 2,457 compatible evaluated joint response-policy pairs generate the convex completion in (22). Proposition 6 gives the same coordinatewise endpoints for this continuum. Its smallest delayed-target margin is 0.002193, its smallest persistent-target margin is 0.002255, and its largest vertex distance is 3.919872 , leaving radius slack 5.59×10^{-5} .

TABLE 8. Continuous response completion at radius 3.9199

Family	Entry witness	Convex margin	Certified lower cells	Recorded $\bar{\delta}^{\text{sign}}$
RANK-RE	3.4280	0.002366	0/34	4.1847
HANK-RE	3.4245	0.002255	0/34	4.2077
RANK-CD	3.3857	0.002193	0/34	4.7428
HANK-CD	3.4089	0.002330	0/34	4.9899

The convex margin is the smallest distance below zero among 34 delayed target coordinates over the family-specific convex response completion. Certified lower cells counts targets with validated global sign-reversal lower bounds. Recorded upper values use 300-quarter evaluated reversals and the baseline metric. The RANK-RE and HANK-CD entries incorporate the LP-directed parameter search, followed by original-protocol input fitting. The HANK-RE entry retains the earlier delay-three witness.

The convex-completion endpoints also give an exact policy-target-scale error budget. If a validation exercise supplies a common coordinatewise additive error bound ε , a delayed

cell retains its negative sign when its recorded upper endpoint plus ε lies below zero. All 34 cells retain the sign for $\varepsilon < 0.0021925988$; the binding coordinate is inflation at horizon sixteen. Table 9 records the full count at six allowances. Solver residuals, response-space approximation errors, and held-out residuals enter this calculation after conversion into the policy-target units by a validated family-specific bound.

TABLE 9. Delayed signs under a deterministic policy-target error allowance

Allowance ε	Output	Inflation	All cells
0.0000	17/17	17/17	34/34
0.0010	17/17	17/17	34/34
0.0020	17/17	17/17	34/34
0.0025	17/17	15/17	32/34
0.0050	17/17	5/17	22/34
0.0100	17/17	0/17	17/34

A cell is counted when its benchmark convex-completion upper endpoint plus ε remains below zero. The exact common budget for all 34 delayed cells is $\varepsilon < 0.0021925988$, attained by inflation at horizon sixteen. The table is a deterministic perturbation calculation. A statistical or numerical error source enters this budget after a valid policy-target-scale bound has been established.

At the binding RANK-CD inflation cell, the independent $T = 300$ solve differs from the response artifact by 2.99×10^{-10} . The $T = 200$ – $T = 300$ change is 7.84×10^{-7} , and the $T = 275$ – $T = 300$ change is 3.73×10^{-8} . Across all 2,457 compatible vertices and 34 targets, 4,914 independent solves give maximum artifact– $T = 300$ and $T = 200$ – $T = 300$ discrepancies of 2.375×10^{-8} and 2.682×10^{-5} , equal to 0.001083 and 1.223 percent of the margin. The maximum relative equilibrium residual is 2.24×10^{-16} . Across 26 distinct maximizing witnesses, the maximum equilibrium condition number is 2.43×10^{12} . The compatible-design local-policy and factor-relation secant maxima are 0.3424 and 0.1658, equal to 156.16 and 75.61 margins. These secant rows serve as approximation diagnostics; a native certificate uses domain-wide nonlinear enclosures.

The baseline-design sensitivity recomputes response distance under three locally available covariance choices and a current-vintage response center. The current-vintage center admits 2,417 draws and preserves the 0.002193 margin. The model-artifact training-only covariance admits 3,215 draws and yields margin 0.001470. The raw rank-50 covariance reaches all-family evaluated coverage at radius 4.2862, above the benchmark, so its benchmark policy set is empty. The online appendix reports the four-row specification path.

The retained-support coordinate boxes are the closed coordinatewise hulls of each family’s 1,000 retained parameter draws. They have five active coordinates in the RE

families and six in the CD families; coordinates with retained standard deviation at most 10^{-8} are fixed. Horizon-300 searches give feasible sign reversals for RANK-RE inflation at delay zero with distance 4.7738, HANK-RE inflation at delay zero with distance 4.2532, and RANK-CD output at delay zero with distance 4.7428. The equilibrium relative residuals are at most 2.03×10^{-17} across these witnesses. Each distance is an upper bound on its family sign-reversal distance.

The HANK-CD search over the retained draws and additional candidate points supplies an output reversal at delay four; reevaluation with the original shock fitting and baseline metric gives distance 4.9899 and target 0.00001217. The RANK-RE witness at 4.1847 and this HANK-CD witness retain positive targets at horizons 250 and 300. Their 300-quarter equilibrium condition numbers are 1.39×10^{13} and 6.68×10^9 , respectively.

The outward-rounded binary64 calculation now constructs a positive-width parameter box with certified response compatibility around each family’s minimum-distance retained draw in the 300-quarter implementation. Compatibility uses the shock path fitted at the center as an existence witness throughout the box. Equilibrium responses, the 34 policy targets, training responses, and covariance whitening are interval-enclosed. Table 10 reports the largest passing grid box in each family. Distance upper bounds range from 3.5580 to 3.7515, below 3.9199, and every target upper bound is negative. The calculation certifies 136 of 136 local compatible family-target cells.

TABLE 10. Validated local native-compatible neighborhoods

Family	Center distance	Halfwidth/span	Neumann upper	Distance upper	Largest target upper
RANK-RE	3.4880	1.2×10^{-14}	0.9236	3.5580	-0.006512
HANK-RE	3.5058	5.0×10^{-14}	0.8842	3.5809	-0.006703
RANK-CD	3.4838	3.0×10^{-11}	0.7740	3.7515	-0.010506
HANK-CD	3.4623	2.0×10^{-13}	0.9254	3.5788	-0.010761

Each box is centered at the family’s minimum-distance retained draw. Halfwidth is a fraction of each active coordinate’s retained-draw span. The center-fitted shock path is held fixed as a feasible input throughout the box. Outward-rounded interval arithmetic encloses the 300-quarter equilibrium, training responses, covariance whitening, and 34 discounted policy targets. A Neumann upper bound below one certifies equilibrium nonsingularity. Each response-distance upper bound lies below 3.9199 and each target upper bound lies below zero. The result covers 136 local family-target cells. The fixed HANK grid, finite-horizon tail, and remainder of the baseline parameter hulls lie outside this certificate.

The outward-rounded boxes establish local continuum statements around four minimum-distance retained points. The finite-support frontiers and these local neighborhoods are reported as distinct domains.

7.7. Delayed and persistent interventions

For each compatible retained point, the model maps a one-quarter 25-basis-point tightening delayed by h quarters into discounted cumulative output and inflation responses. The evaluated set and its convex response completion share the endpoints in Table 11.

TABLE 11. Cumulative responses to delayed 25-basis-point tightenings

Delay h	Output endpoint interval	Inflation endpoint interval
0	$[-0.110, -0.031]$	$[-0.019, -0.002]$
4	$[-0.179, -0.054]$	$[-0.032, -0.005]$
8	$[-0.211, -0.063]$	$[-0.037, -0.005]$
12	$[-0.183, -0.044]$	$[-0.033, -0.004]$
16	$[-0.120, -0.011]$	$[-0.026, -0.002]$

Entries are the union endpoints across 2,457 compatible retained baseline evaluations from four model families. Proposition 6 gives the same endpoints for their continuous joint response–policy convex completion. Targets discount quarterly responses at $\beta = 0.99$.

Both endpoints are negative at every delay from zero through sixteen quarters. Across the 34 target–delay cells, output endpoints range from -0.211 to -0.011 and inflation endpoints range from -0.037 to -0.002 . At delay eight, output lies in $[-0.211, -0.063]$ and inflation lies in $[-0.037, -0.005]$. The smallest margin occurs for inflation at delay sixteen. The output profile reaches its largest absolute lower endpoint near delay eight.

Persistent interventions combine the delayed basis policies according to $u_t = 0.25\rho^t$. Table 12 reports the corresponding endpoints.

TABLE 12. Cumulative responses to persistent tightening paths

Persistence ρ	Output endpoint interval	Inflation endpoint interval
0.00	$[-0.110, -0.031]$	$[-0.019, -0.002]$
0.50	$[-0.254, -0.074]$	$[-0.045, -0.006]$
0.75	$[-0.594, -0.184]$	$[-0.105, -0.016]$
0.90	$[-1.383, -0.438]$	$[-0.240, -0.038]$

The path is $u_t = 0.25\rho^t$ for $t = 0, \dots, 16$. Linearity combines the compatible delayed-intervention responses before endpoints are taken. The four rows use the same 2,457 retained evaluations and continuous convex completion as Table 11.

At $\rho = 0.50$, output lies in $[-0.254, -0.074]$ and inflation in $[-0.045, -0.006]$. At $\rho = 0.90$, the intervals are $[-1.383, -0.438]$ and $[-0.240, -0.038]$. These endpoints apply to the compatible retained evaluations and their convex response completion. The SPF response supplies an observable counterpart at horizons where a persistent intervention continues to load on the economy.

The downstream rule calculation is reported in Online Appendix E as a feasibility diagnostic. Its median policy peaks are 39.95–116.41 annualized percentage points, with maximum 218.45.

7.8. Weak-rank inference diagnostics

The four mode response Jacobians have ranks 7, 7, 9, and 9, equal to their parameter dimensions. Local policy visibility is positive in all 136 family–target–horizon cells, with $\kappa_{f,p}(h)$ from 1.417 to 98.064. These calculations establish local tangent visibility at the family modes.

The weak-rank Gaussian experiment contains 648 cells and 3.24 million replications. Table 13 reports the 324 cells with $c \leq 1$ and positive target loading, together with the 27 fixed-positive-singular-value cells.

TABLE 13. Weak-rank inference: size, power, and confidence-set geometry

Procedure and design	Rejection	Power	Unbounded	Empty	Infinite median
Projection, weak sequence	0.146	0.284	74.318	0.053	243
Target-constrained oracle, weak sequence	4.996	6.185	67.306	2.268	243
Quotient, fixed positive singular value	5.031	75.024	0.000	2.042	0

The weak rows contain 324 cells; the regular row contains 27. Rejection ranges in row order are [0.020, 0.340], [4.220, 5.800], and [4.380, 5.740] percent. Rejection, power against a 0.25 target shift, unbounded-set frequency, and empty-set frequency are percentages. The target-constrained row uses known covariance and analytic profiling.

In the weak cells, the profiled null statistic has a χ^2_3 law. Least-favorable projection uses the $\chi^2_{8,95}$ critical value 15.507. Its analytic rejection probability is 0.1431 percent, close to the simulated mean 0.1463 percent. Mean power against a 0.25 shift is 0.2844 percent, mean unbounded-set frequency is 74.32 percent, and 243 of 324 cells have infinite median length. These quantities measure the inferential cost of the rank-uniform reference. Dufour (1997) establishes that valid confidence sets for unbounded weakly identified targets can require positive probability of unboundedness. The particular excess size conservatism here comes from using eight reference degrees of freedom for a three-dimensional profiled Gaussian statistic. The reported calculations measure both effects through rejection, power, and set length.

For positive target loading, the target-null level set removes the weak direction and the augmented null Jacobian has singular value $\sqrt{\gamma^2 + c^2/T} \geq \gamma$. The target-constrained quotient oracle has mean rejection 4.996 percent and mean power 6.185 percent in the weak cells; its mean unbounded-set frequency remains 67.31 percent. In the regular supplement,

rejection ranges from 4.38 to 5.74 percent, mean power is 75.02 percent, and every reported set is bounded. Application of the target-constrained refinement uses the diagnostics following Theorem 8.

7.9. Primitive response uncertainty and re-estimation

The raw-data reconstruction uses 275 dated quarters from 1954Q3 through 2023Q1. It refits the Hamilton regression on its full lead-lag footprint, estimates responses over 1969Q1–2006Q4, and recomputes each shock’s 25-basis-point peak normalization. Calendar weights enter Hamilton dependent-variable dates and VAR residual or LP forecast-origin dates. Numerical derivatives of this two-step estimator supply a 275×150 primitive influence array. The dependence calculation applies a Bartlett HAC sandwich on that common calendar with bandwidths 8, 16, and 28. Recursive RR impact restrictions and fixed peak-normalized rate ordinates have zero derivative.

The VAR point reproduces the baseline normalized response within 1.28×10^{-11} . Its primitive derivative has numerical rank 40. The selected 50-coordinate training covariance is singular, so the positive-definite distance in (6) requires a different response coordinate system or a singular-experiment extension. The LP estimator uses three lags and horizon-specific forecast-origin samples. Its 50-coordinate training covariance is positive definite under all three reported bandwidths. Halving the case-weight differentiation step changes the derivative arrays by relative norms below 1.22×10^{-7} , with zero peak-index switches.

TABLE 14. Response estimator and fixed-input compatibility

Experiment	Coordinates	Entry	Reversal	Count
Baseline taper	52	3.51	4.36	2,457
VAR, HAC 28	50	–	–	–
LP, HAC 8	50	27.22	29.49	0
LP, HAC 16	50	36.11	38.27	0
LP, HAC 28	50	46.43	49.82	0

Entry and reversal are baseline-design radii. Count is the number of evaluations at radius 3.9199. New rows use the original parameter support and fitted inputs, rescaled to the estimator’s peak normalization. The VAR derivative has rank 40 across 150 primitive responses; its 50-coordinate training covariance is singular, so the full-rank distance is withheld. LP uses three lags. All HAC rows use the 275-quarter calendar and refit preprocessing and normalization.

Table 14 keeps the original structural parameters and fitted shock inputs fixed, converting their normalization to the new estimator’s scale. Every LP row has an empty compatible set at 3.9199. Family entry occurs at radii 27.22, 36.11, and 46.43 for bandwidths 8, 16, and 28. The corresponding evaluated reversal distances are 29.49, 38.27, and 49.82.

These rows measure estimator and covariance sensitivity at the original fits. The following refitting experiment varies both structural parameters and shock paths.

The primitive covariance estimates use the stated raw-data vintage and shock-series construction and provide the empirical metric for the subsequent structural refitting exercise.

7.10. Structural refitting to the LP experiment

Let $G_f(\theta)q$ collect the 150 model responses, where q stacks twenty-five coefficients for each shock. For response coordinates A , the refitting criterion is

$$(61) \quad \tilde{d}_{f,A,b}(\theta) = \min_{q \in \mathcal{Q}_f(\theta)} \left\| \hat{\Sigma}_{A,b}^{-1/2} ([G_f(\theta)q]_A - \hat{s}_A^{LP}) \right\|_2.$$

Both shock-path profiling and parameter search minimize this same criterion. The set $\mathcal{Q}_f(\theta)$ fixes the AD rate response at quarter two and the RR rate response at quarter one to 0.0625 quarterly percentage points, fixes RR output and inflation impact responses to zero, and bounds the absolute rate response at every quarter zero–twenty-four by 0.0625. These restrictions preserve the declared recursive zeros and peak-normalization cell. The earlier mixed-objective search profiled inputs on 146 coordinates and searched parameters using a 50-coordinate subdistance, with two peak equalities. Equation (61) records the common criterion and the additional normalization restrictions used here.

The parameter search uses the retained-evaluation coordinate hulls and fixes inactive parameters to the first retained evaluation. These are computational domains defined by the retained draws. The economic-domain and household-grid declarations retain their separate certification requirements. The principal LP experiment uses the fifty stochastic output and inflation coordinates through quarter twelve. The 146-coordinate experiment also fits the rate path and responses through quarter twenty-four, omitting four deterministic coordinates. Each experiment profiles its own shock paths. Policy targets retain the original announced 25-basis-point wedge definition.

Fifty-two parameter-search runs generate 102 distinct parameter evaluations across the four families. Reprofiled every point under both coordinate choices and all three bandwidths gives 612 constrained fits on a common parameter support. Every fit satisfies the numerical inner-solver and normalization checks. The largest relative convex-profile gap is 3.74×10^{-10} , the largest normalization equality error is 3.56×10^{-15} , and the largest rate-bound violation is 2.64×10^{-15} . The parameter searches supply evaluated candidates; their global optimality remains a separate question.

TABLE 15. LP-refitted distances on a common evaluated parameter support

Family	Entry, 50	Reversal, 50	Entry, 146	Reversal, 146	Bounds
RANK-RE	23.2840	32.2795	274.28	352.71	30
HANK-RE	18.2001	25.8588	323.16	372.30	32
RANK-CD	22.3531	27.1542	240.11	290.05	39
HANK-CD	17.0400	17.1449	316.26	559.40	29
Family union	23.2840	17.1449	323.16	290.05	–

The covariance uses Bartlett bandwidth 28. Every row profiles both 25-quarter shock paths with the same objective used for parameter search. The 102 distinct parameter points contain 25 RANK-RE, 28 HANK-RE, 25 RANK-CD and 24 HANK-CD points. Each point is reprofiled under both experiments; the 50-coordinate experiment uses output and inflation at horizons zero–twelve, and the 146-coordinate experiment adds the other stochastic primitive responses. Rate peaks and recursive zeros are imposed as in (61). Bounds counts active normalized-rate bounds among 50 ordinates at the 50-coordinate entry point, including the two peak anchors. Union entry is the largest family minimum; union reversal is the smallest nonnegative-target distance. These are evaluated-support quantities on the declared parameter support.

For the fifty-coordinate HAC-28 experiment, family entry occurs at 23.2840 and the first evaluated reversal at 17.1449. Thus reversal precedes all-family entry, and the common-negative-sign interval is empty on this support. At radius 3.9199, the evaluated compatible set is empty. HAC bandwidths 8 and 16 give entry–reversal pairs (14.3999, 11.1563) and (17.5390, 14.2231). The same ordering appears in each row. These are refitted finite-support findings on the declared parameter support.

The first HAC-28 LP reversal is HANK-CD output at delay four. Consumption contributes -0.00709636 and investment contributes 0.00710854 , giving target 0.00001217 . At a 250-quarter solve it remains positive at 0.00001102 . Its baseline-metric distance with the original shock-fitting protocol is 4.9899 . This comparison attaches the same structural counterfactual to two specified response experiments.

Table 16 attributes the baseline-LP change on declared supports. Panel A holds the 102 LP-profiled fitted response vectors and policy targets fixed. Replacing baseline response dynamics moves the gap from -0.0683 to -0.9548 ; applying the LP peak normalization moves it to -1.0918 ; applying the LP HAC-28 covariance moves it to -6.1391 . Panel B separates parameter support from shock fitting under the LP metric. The 4,000 baseline points and fitted inputs give gap 3.3811 . Evaluating the original shock-fitting protocol on the 102 LP-search parameter points gives 6.2580 . Applying the LP shock-profile criterion on those same points gives -6.1391 . The fitted shock path and covariance geometry account for the sign-ordering change in this calculation.

TABLE 16. Decomposition of the baseline–LP frontier change

Panel A. Center, normalization, and covariance on fixed LP-profiled responses

Response experiment	Entry	Reversal	Gap
Baseline center and covariance	5.5149	5.4466	-0.0683
LP dynamics, baseline normalization	11.2252	10.2704	-0.9548
LP center, baseline covariance	11.5117	10.4199	-1.0918
LP center and HAC 28 covariance	23.2840	17.1449	-6.1391

Panel B. Parameter support and shock fitting under the LP HAC-28 experiment

Structural calculation	Points	Entry	Reversal	Gap
Baseline 4,000 points and fitted inputs	4,000	46.4344	49.8155	3.3811
LP-search points, original shock fitting	102	41.0373	47.2953	6.2580
LP-search points, LP shock profiling	102	23.2840	17.1449	-6.1391

Panel A uses the same 102 LP-profiled fitted response vectors and policy targets. The first row uses the baseline center and active 50×50 covariance. The second row replaces the response dynamics while retaining the baseline peak-normalization multipliers; the third applies the LP peak normalization; the fourth applies the LP HAC-28 covariance. Panel B starts from the 4,000 baseline evaluations and fitted inputs, then evaluates the original full-response shock-fitting protocol on the 102 LP-search parameter points, and finally applies the LP shock-profile criterion on those points. Each row is an evaluated-support calculation.

On 146 coordinates, the HAC-28 entry radius is 323.1614 and the reversal radius is 290.0486. Its covariance condition number is 3.24×10^9 , compared with 1.62×10^5 for fifty coordinates. These large discrepancies accompany the additional rate and long-horizon coordinates and their estimated covariance geometry. Their interpretation is a profiled response-distance diagnostic; sampling rejection probabilities require the generated-response distribution and constrained inversion. Online Appendix E reports bandwidth paths, active rate bounds, and horizon checks.

7.11. Sampling variation in the LP gap

Table 17 reports family entry, first reversal, and their gap for the baseline experiment and the three fifty-coordinate LP experiments. Proposition 5 maps these experiment-specific rows into the collection criterion.

TABLE 17. Experiment-indexed family-entry and sign-reversal gaps

Experiment	Entry	Reversal	Gap	95% gap range	Pr(gap > 0) (%)
Baseline covariance	3.5058	4.3567	0.8509	–	–
LP, HAC 8	14.3999	11.1563	-3.2435	[-5.1655, -1.2974]	0.033
LP, HAC 16	17.5390	14.2231	-3.3159	[-5.2075, -1.3484]	0.003
LP, HAC 28	23.2840	17.1449	-6.1391	[-7.8345, -4.2955]	0.000

The gap is $c_{\text{sign}} - c_{\text{entry}}$. The baseline row is a point comparison under its baseline covariance. LP rows use 100,000 Gaussian perturbations of the declared raw-data response estimator and report sampling ranges. Each profiled distance is linearized by its envelope derivative, and the family minima, all-family maximum, and reversal minimum are recomputed in every draw across the 102-point support. The calculation permits binding evaluations to change. Its scope conditions on the evaluated structural support, policy targets, covariance estimator, and active profiling regime.

For the LP rows, the empirical center receives Gaussian perturbations from the raw-data covariance. The envelope derivative of each profiled distance supplies its local response to the perturbation. Family minima, the all-family maximum, and the reversal minimum are recomputed in every draw across all 102 evaluated points. Binding evaluations change across draws: under HAC 28, the two most frequent entry evaluations account for 71.55 percent of draws, and three HANK-CD reversal evaluations account for the reversal minimum. The directional calculation retains these switches. The 95-percent gap ranges lie below zero for all three bandwidths. These ranges condition on the evaluated structural support, fixed policy targets, covariance estimator, and active profiling regime; parameter-domain optimization and full generated-response inference remain separate objects.

The selected calendar bootstrap uses 28-quarter circular blocks and adaptive peak normalization. Each of its 1,999 draws reruns Hamilton detrending, lag-augmented LP estimation, impact identification, response normalization, and all 102 constrained shock-path profiles, then recomputes the family-entry and reversal binders. Table 18 reports the resulting distribution. The point gap is -6.1391 , the bootstrap median is -4.9865 , and the 95-percent percentile interval is $[-10.2187, 2.2214]$. The gap is positive in 6.353 percent of draws. The peak quarter changes in 1,421 draws. All draws produce a frontier, and the maximum width of the propagated numerical gap envelope is 3.01×10^{-5} . The HAC metric, 102-point structural support, and policy targets remain fixed across draws.

TABLE 18. Full-reprofile calendar bootstrap for the selected LP gap

Block	Point	2.5%	Median	97.5%	$\Pr(\Delta > 0)$ (%)	Peak switches
28	-6.1391	-10.2187	-4.9865	2.2214	6.353	1,421

The table reports percentile quantiles from 1,999 circular calendar block-multiplier draws with adaptive peak normalization. Each draw reruns Hamilton detrending, lag-augmented local projections, Cholesky impact identification, and all constrained shock-path profiles, then recomputes the family-entry and reversal binders over the 102-point support. The HAC-28 metric, structural support, and policy targets remain fixed. All draws produce a frontier. The maximum numerical gap-envelope width is 3.01×10^{-5} .

7.12. Magnitude thresholds

For a cumulative-response threshold $\Delta \geq 0$, define $c_j^-(\Delta) = \inf\{d(M) : T_j(M) \geq -\Delta\}$. It measures breakdown of the conclusion $T_j < -\Delta$. The sign frontier uses $\Delta = 0$. Table 19 reports a sensitivity grid in the target's discounted units. At $\Delta = 0.001$, all seventeen inflation cells satisfy the threshold at the benchmark and the first evaluated crossing is 4.0319. At $\Delta = 0.005$, five inflation cells satisfy it and the crossing is 3.5213. Output satisfies all four reported thresholds in every delayed cell at the benchmark.

TABLE 19. Evaluated frontiers for sign and magnitude conclusions

Δ	Output radius	Inflation radius	Output cells	Inflation cells
0.000	–	4.3567	17	17
0.001	–	4.0319	17	17
0.005	–	3.5213	17	5
0.010	4.1257	3.4795	17	0

The conclusion is $T_j < -\Delta$; a reversing evaluation has $T_j \geq -\Delta$. Thresholds use the discounted cumulative units of the policy target and provide a sensitivity grid. Cell counts are out of seventeen per outcome at radius 3.9199. A dash means the evaluated support contains zero reversing evaluations.

For a positive reversal of size $\delta \geq 0$, define $c_j^+(\delta) = \inf\{d(M) : \max_h T_{j,h}(M) \geq \delta\}$. Table 20 applies this object to the common 102-point LP support under HAC 28. The output and inflation sign frontiers are 17.1449 and 25.8588. At $\delta = 0.0001$, they move to 27.3523 and 36.8739. The largest positive targets on the support are 0.000284 and 0.000814, so the support contains zero qualifying evaluations for $\delta \in \{0.001, 0.005, 0.010\}$. These thresholds are absent from the evaluated support.

TABLE 20. Positive-reversal thresholds in the LP-refitted support

δ	Output radius	Output points	Inflation radius	Inflation points
0.0000	17.1449	5	25.8588	12
0.0001	27.3523	1	36.8739	2
0.0010	–	0	–	0
0.0050	–	0	–	0
0.0100	–	0	–	0

The frontier is $c_j^+(\delta) = \inf\{d(M) : \max_h T_{j,h}(M) \geq \delta\}$ on the common 102-point LP support under HAC bandwidth 28. The zero row is the sign frontier. Positive thresholds distinguish a numerical crossing from a reversal of stated size. A dash records zero qualifying evaluations on this support and carries support-level information only. Targets use the discounted cumulative units of the policy exercise.

7.13. Additional response checks

Family-held-out prediction evaluates 68 family–horizon cells, with minimum predictive coverage 0.863 and zero five-percent generated-loss rejections. Treasury-yield, SPF, funding-spread, QFR, SCF, and CEX estimates provide additional transmission evidence. These response checks are reported on their own empirical scales and complement the training-response frontier.

Under the baseline response experiment, family entry and common negative signs hold on the compatible retained design and its convex response–policy completion. Four outward-rounded parameter neighborhoods also remain inside the benchmark with 136 negative target bounds. On the refitted LP support, the first sign crossing precedes all-family entry and every positive target is below 0.001. The full-reprofile calendar interval spans zero, with 6.353 percent of draws producing a positive gap.

8. Conclusion

The family-entry and reversal radii make cross-model counterfactual agreement specific to a causal-response experiment. A positive gap identifies a response neighborhood that contains every declared family while preserving the policy conclusion. Evidence increments show which observed responses move the reversal boundary. Empirical-fiber factorization gives the corresponding sufficiency condition, and generated-response inference carries sampling variation into the frontier.

The monetary application gives different orderings across two causal-response experiments. The baseline experiment has entry radius 3.5058 and first evaluated reversal 4.3567. RR responses and the seventh training quarter account for the separation. The local-projection experiment gives negative gaps under three HAC bandwidths after struc-

tural parameters and shock paths are refitted. On a common parameter support, changing shock profiling moves the HAC-28 gap from 6.2580 to -6.1391 ; covariance replacement accounts for most of the remaining fixed-response movement. The positive LP crossings are smaller than 0.001 in discounted cumulative units. A CMR calibration point links an output reversal to the timing of consumption and investment and to the RR evidence that excludes it.

The sampling and domain calculations define the scope of each conclusion. The full-reprofile calendar bootstrap gives a 95-percent LP-gap interval of $[-10.2187, 2.2214]$ and a 6.35-percent positive-gap frequency. Outward-rounded calculations certify four positive-width parameter neighborhoods inside the benchmark with 136 negative target bounds. The reported frontiers apply to their named response experiments, evaluated supports, convex response-policy completions, and local certified neighborhoods. Within each of these objects, the evidence needed for a monetary counterfactual is measured by the response distance at which structural families first disagree.

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